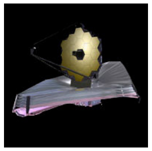

10394 - Gaseous Water in Warm Exo-Asteroid Belts: Observational Tests of a Sublimation-Driven Hydration Mechanism

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
Observation Folder				
	1	HD 1466 MRS	MIRI Medium Resolution Spectroscopy	(3) HD-1466
	2	HD 1466 MRS BACK GROUND	MIRI Medium Resolution Spectroscopy	(4) HD-1466-background

ABSTRACT

In the Solar System, water delivery to terrestrial planets is hypothesized to have occurred via asteroidal impacts. We propose a novel hydration mechanism for exoplanetary systems (which may also work in the Solar System): sublimation-driven water transfer from warm exo-asteroid belts. The young system HD 1466 hosts a massive warm belt where water ice should sublimate efficiently, releasing H2O gas. Kral+24 developed a new model that tracks ice sublimation, viscous gas evolution, and planetary accretion, predicting inward viscous spreading of water vapour via a circumstellar disc that enables efficient terrestrial planet hydration.

This proposal aims to achieve the first ever detection of gaseous water in a mature planetary system (post protoplanetary disc phase), targeting lines between 5 and 20 mic that are very sensitive to low masses of water vapour and OH gas. Leveraging JWST's proven capability to map water in protoplanetary discs (e.g., Sz 114, Xie+23), we ran a set of new simulations confirming detectability in HD 1466 even under conservative assumptions. A detection in such an exo-asteroid belt would establish a universal pathway for hydrating terrestrial planets without impacts (i.e. the process would be more predictable and less contingent). Success here would motivate systematic surveys to quantify aqueous delivery in exo-systems and reshape our understanding of volatile evolution in inner planetary habitability zones.

OBSERVING DESCRIPTION

We propose to observe the dust and gas in HD 1466 using MIRI/MRS. The gas and dust in this system are located a few au away, and they will not be spatially resolved. The lines we are targeting (water and OH) should be broadened by Keplerian shear at more than 20-100 km/s and will be barely resolved spectrally, but blending will be avoided thanks to the medium resolution of the MRS.

In order to obtain a complete spectrum from 5 to 28 microns, with sufficient spectral resolution to detect OH, ortho and para water lines, we will observe this system with MIRI/MRS in all 4 channels. The MIRI MRS observations will produce a spectrum in each pixel with a spectral resolution of between 1300 and 3700, depending on the wavelength, which is essential for achieving all our scientific objectives. This is the first time that the MRS would be used to characterise an exo-asteroid belt very similar to that in our own solar system, a true asteroid belt analogue. The goal is to detect a disc of water vapour in this system, showing that water delivery onto planets may not come from (exo-)asteroidal impacts but rather disc spreading. We will also detect the dust disc in this system with S/N > 10 between 10 and 20 mic.

Methodology: As the disc is not resolved in the central pixel, the stellar flux will be mixed with the disc flux. We will therefore eliminate the stellar component by modelling it, as was done in the past with Spitzer for similar spectra (Beichman+06, Lisse+07) with a precision of 1-2% on the stellar spectrum. To get a usable disc spectrum, an SNR of at least 300 must be obtained in each spectral channel for the stellar + disc observations (private comm. with GTO MINDS on sources and equivalent programmes). This SNR can be obtained with an observing time of 30 min for channels 1, 2, 3 and 4. We will also remove the disc continuum to access the lines we are after. To complete our setup, we need to acquire the target and the dither pattern of the point source (4 points) in order to calibrate the fringes (Gasman+23). We also need a background observation equivalent to a dither point. In addition, we will use the spectra 515 Athalia and 526 Jena obtained by the GO 1549 programme in cycle 1 to calibrate the spectrum of HD 1466. The latter are bright main-belt asteroids (featureless) observed to calibrate the equivalent MRS observation of circumstellar discs as shown in Xie+23.

The total duration of the programme is 5.0 hours.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(3)	HD-1466	RA: 00 18 26.1236 (4.6088483d) Dec: -63 28 38.98 (-63.47749d) Equinox: J2000	Proper Motion RA: 90.053 mas/yr Proper Motion Dec: -59.20699998114287 mas/yr Parallax: 0.023355499999999998" Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Debris disks, G stars]			
(4)	HD-1466-background	RA: 00 18 13.0000 (4.5541667d) Dec: -63 29 10.00 (-63.48611d) Equinox: J2000	Proper Motion RA: 90.053 mas/yr Proper Motion Dec: -59.20699998114287 mas/yr Parallax: 0.023355499999999998" Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Calibration Description=[Telescope/sky background]			

Observation

Proposal 10394, Observation 1: HD 1466 MRS

Fri Mar 13 19:06:49 GMT 2026

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Background Observations:[HD 1466 MRS BACKGROUND (Obs 2)]

Diagnostics

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(3)	HD-1466	RA: 00 18 26.1236 (4.6088483d) Dec: -63 28 38.98 (-63.47749d) Equinox: J2000	Proper Motion RA: 90.053 mas/yr Proper Motion Dec: -59.20699998114287 mas/yr Parallax: 0.023355499999999998" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.
Category=Star
Description=[Debris disks, G stars]

Acquisition

#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	SAME	FND	FAST	10	1	1	27.75	217981.13

Template

Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
All MRS	NO	FULL	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	POINT SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	SHORT(A)	MRSLONG		FASTR1	28	5	1	Dither 1	4	20	1598.423	273909
1	SHORT(A)	MRSSHORT		FASTR1	28	5	1	Dither 1	4	20	1598.423	
2	MEDIUM(B)	MRSLONG		FASTR1	28	5	1	Dither 1	4	20	1598.423	
2	MEDIUM(B)	MRSSHORT		FASTR1	28	5	1	Dither 1	4	20	1598.423	
3	LONG(C)	MRSLONG		FASTR1	28	5	1	Dither 1	4	20	1598.423	
3	LONG(C)	MRSSHORT		FASTR1	28	5	1	Dither 1	4	20	1598.423	

Special Requirements	Sequence Observations 1, 2, Non-interruptible

Observation

Proposal 10394, Observation 2: HD 1466 MRS BACKGROUND

Fri Mar 13 19:06:49 GMT 2026

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Background Observation For: [HD 1466 MRS (Obs 1)]

Diagnostics

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(4)	HD-1466-background	RA: 00 18 13.0000 (4.5541667d) Dec: -63 29 10.00 (-63.48611d) Equinox: J2000	Proper Motion RA: 90.053 mas/yr Proper Motion Dec: -59.20699998114287 mas/yr Parallax: 0.023355499999999998" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.
Category=Calibration
Description=[Telescope/sky background]

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
FND	All MRS	NO	FULL	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	POINT SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	SHORT(A)	MRSLONG		FASTR1	28	5	1	Dither 1	4	20	1598.423	
1	SHORT(A)	MRSSHORT		FASTR1	28	5	1	Dither 1	4	20	1598.423	
2	MEDIUM(B)	MRSLONG		FASTR1	28	5	1	Dither 1	4	20	1598.423	
2	MEDIUM(B)	MRSSHORT		FASTR1	28	5	1	Dither 1	4	20	1598.423	
3	LONG(C)	MRSLONG		FASTR1	28	5	1	Dither 1	4	20	1598.423	
3	LONG(C)	MRSSHORT		FASTR1	28	5	1	Dither 1	4	20	1598.423	

Special Requirements	
	Sequence Observations 1, 2, Non-interruptible